

VisionInspect AI

Manufacturing Defect Detection & Quality Inspection System

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Milestone-2

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1. Introduction

VisionInspect AI is an AI-powered manufacturing quality inspection system designed to automate the detection of product defects using computer vision and deep learning. The application integrates modern web technologies with deep learning models to assist quality engineers and factory supervisors in identifying manufacturing defects accurately and efficiently.

During Milestone 2, the project evolved from independently developed modules into a unified inspection platform. The backend services, augmentation pipeline, preprocessing modules, AI inspection system, and frontend dashboard were successfully integrated into a single working application capable of performing real-time inspection of uploaded manufacturing images.

Major additions in this milestone included dataset augmentation, API integration, frontend-backend communication, defect visualization, anomaly heatmaps, YOLO-based product localization, autoencoder reconstruction, severity analysis, and unified project architecture.

2. Milestone 2 Objectives

- Implement dataset augmentation to improve model robustness.
- Develop preprocessing APIs for manufacturing images.
- Build modular backend services using FastAPI.
- Integrate backend modules into a unified application.
- Connect the frontend dashboard with backend APIs.
- Replace simulated inspection results with real AI predictions.
- Display AI-generated visualizations including heatmaps and reconstructed images.
- Support category-wise inspection for multiple manufacturing products.

3. Dataset Augmentation

Deep learning models require large and diverse datasets to achieve high accuracy and strong generalization.

Manufacturing datasets are often limited in size, making augmentation an essential preprocessing step.

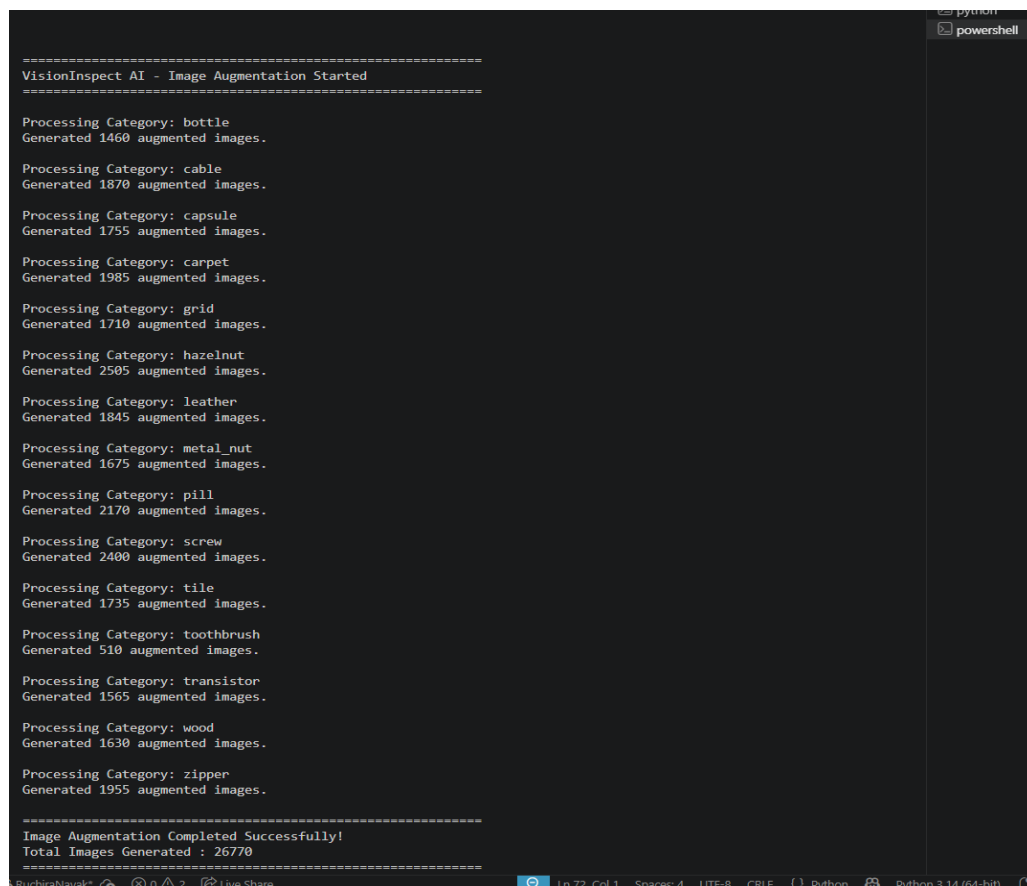
To improve dataset diversity, an image augmentation module was implemented. This module generates multiple variations of the original images while preserving their semantic information.

The augmentation pipeline includes the following transformations:

- Image Rotation
- Horizontal Flip
- Vertical Flip
- Brightness Adjustment
- Contrast Enhancement
- Noise Injection
- Image Normalization

These augmentations increase the effective dataset size, reduce overfitting, and improve the robustness of the anomaly detection model against real-world manufacturing variations.

The augmentation functionality was exposed through a dedicated FastAPI endpoint, allowing images to be processed directly through the backend.



```
=====
VisionInspect AI - Image Augmentation Started
=====

Processing Category: bottle
Generated 1460 augmented images.

Processing Category: cable
Generated 1870 augmented images.

Processing Category: capsule
Generated 1755 augmented images.

Processing Category: carpet
Generated 1985 augmented images.

Processing Category: grid
Generated 1710 augmented images.

Processing Category: hazelnut
Generated 2505 augmented images.

Processing Category: leather
Generated 1845 augmented images.

Processing Category: metal_nut
Generated 1675 augmented images.

Processing Category: pill
Generated 2170 augmented images.

Processing Category: screw
Generated 2400 augmented images.

Processing Category: tile
Generated 1735 augmented images.

Processing Category: toothbrush
Generated 510 augmented images.

Processing Category: transistor
Generated 1565 augmented images.

Processing Category: wood
Generated 1630 augmented images.

Processing Category: zipper
Generated 1955 augmented images.

=====
Image Augmentation Completed Successfully!
Total Images Generated : 26770
=====
```

POST /augment Augment Dataset

Parameters

No parameters

Execute

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

"string"

4. System Architecture

-Frontend Layer

Developed using:

- Next.js
- React
- Tailwind CSS

The frontend provides:

- Login Interface
- Inspection Dashboard
- Image Upload Panel
- Inspection Results

- Heatmap Visualization
- Supervisor Dashboard

-Backend Layer

Implemented using FastAPI.

Responsible for:

- Image Upload
- Dataset Loading
- Image Preprocessing
- Dataset Augmentation
- Dataset Statistics
- API Routing
- Communication with AI Models

-AI Layer

The AI inspection pipeline performs:

- YOLO Product Detection
- Product Cropping
- Autoencoder Reconstruction
- Defect Classification
- Severity Analysis
- Heatmap Generation

-Data Layer

Stores:

- Inspection History
- Analytics
- Quality Reports
- Production Reports

5. Backend Development & Integration

One of the major contributions during Milestone 2 was developing and integrating backend services into a unified FastAPI application.

Initially, preprocessing, augmentation, dataset handling, and AI inspection modules existed as independent components. These services were integrated into a single backend capable of serving both image-processing APIs and AI inference endpoints.

The backend also communicates seamlessly with the frontend dashboard, enabling real-time manufacturing inspection directly from the web interface.

Backend Modules

The integrated backend includes:

- Upload API
- Dataset Loader

- Image Preprocessing
- Dataset Augmentation
- Dataset Statistics
- AI Prediction
- Quality Analysis
- Batch Inspection
- Inspection History
- Analytics
- Production Report Generation

FastAPI Integration

Independent backend modules were combined into a single FastAPI application using modular routers.

Integrated routers include:

- Upload Router
- Dataset Router
- Preprocessing Router
- Augmentation Router
- Statistics Router

This modular architecture improves maintainability while exposing all APIs through a single Swagger interface.

Frontend Integration

The frontend inspection dashboard was connected with the backend APIs.

Instead of simulated inspection data, the dashboard now uploads manufacturing images directly to the backend using multipart form requests.

The backend processes the uploaded image through the AI pipeline and returns:

- Product Category
- Prediction
- Confidence Score
- Severity Score
- YOLO Crop
- Autoencoder Reconstruction
- Heatmap
- Defect Localization Overlay

These outputs are displayed directly inside the inspection dashboard.

Category Support

Support was extended from a single product category to dynamic category selection.

The dashboard now allows users to inspect multiple manufacturing product categories through a dropdown

interface. This architecture enables future addition of new product categories without requiring frontend modifications.

VisionInspect AI - Anomaly Detection API 1.1.0 OAS 3.1

/openapi.json

API for detecting manufacturing defects using Unsupervised Convolutional Autoencoders.

default			^
POST	/upload	Upload Image	▼
POST	/preprocess	Preprocess	▼
GET	/dataset	Load Dataset	▼
POST	/augment	Augment Dataset	▼
GET	/statistics	Statistics	▼
GET	/	Get Dashboard	▼
GET	/status	Get Status	▼
POST	/predict	Predict	▼
POST	/quality-check	Quality Check	▼
POST	/batch-predict	Batch Predict	▼
GET	/history	Get History	▼
GET	/analytics	Get Analytics	▼
GET	/analytics/trends	Get Analytics Trends	▼
GET	/analytics/risk-assessment	Get Risk Assessment	▼
GET	/reports/production	Get Production Report	▼
GET	/report/{inspection_id}	Get Report	▼
POST	/reload	Reload Model Weights	▼

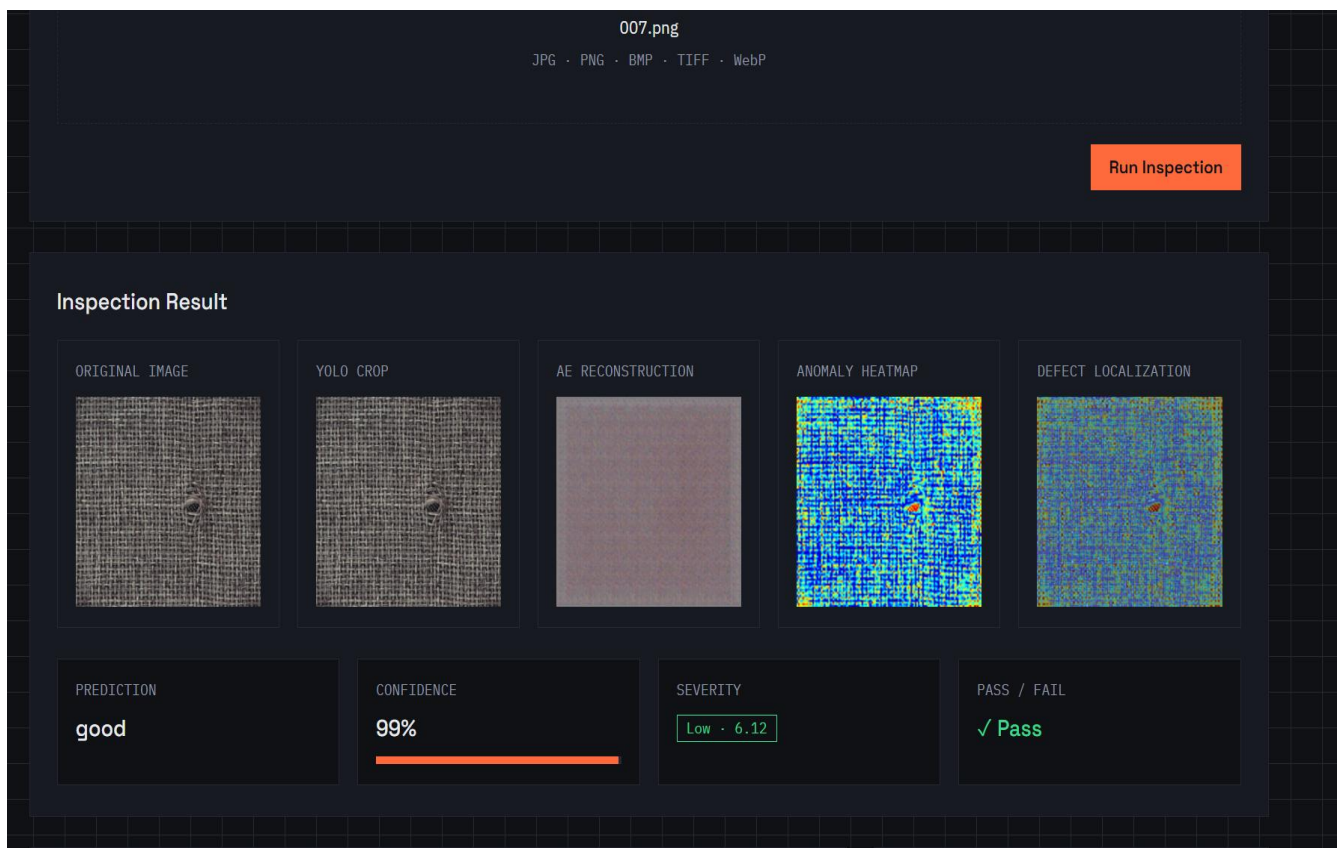
6. AI Inspection Pipeline

The complete inspection workflow follows the sequence below:

1. Image Upload
2. YOLO Product Detection

3. Product Cropping
4. Image Preprocessing
5. Autoencoder Reconstruction
6. Anomaly Heatmap Generation
7. Defect Classification
8. Severity Analysis
9. Inspection Report Generation

This sequential pipeline enables accurate localization, visualization, and classification of manufacturing defects while providing meaningful quality assessment metrics.



7. API Documentation

The following APIs were successfully implemented and integrated.

API Endpoint	Purpose
/upload	Upload manufacturing image
/dataset	Load dataset information
/preprocess	Image preprocessing
/augmentation	Dataset augmentation
/statistics	Dataset statistics
/predict	Defect prediction
/quality-check	Image quality analysis
/batch-predict	Batch inspection
/history	Inspection history
/analytics	Inspection analytics
/analytics/trends	Defect trend analysis
/analytics/risk-assessment	Risk analysis
/reports/production	Production report
/report/{inspection_id}	Individual inspection report
/reload	Reload model weight

8. Testing Results

The integrated application was tested using both Swagger UI and the frontend dashboard.

The following functionalities were successfully verified:

- Image Upload
- Dataset Loading
- Image Preprocessing
- Dataset Augmentation
- Dataset Statistics
- YOLO Product Detection
- Autoencoder Reconstruction
- Heatmap Generation
- Defect Classification
- Severity Calculation
- Dashboard Integration
- Analytics APIs
- Production Reports
- Inspection History

The system successfully processed manufacturing images and generated complete inspection reports without major runtime issues.

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/predict?category=Carpet&enable_yolo=true' \
  -H 'accept: application/json' \
  -H 'Content-Type: multipart/form-data' \
  -F 'file=@007.png;type=image/png'
```

Request URL

http://127.0.0.1:8000/predict?category=Carpet&enable_yolo=true

Server response

Code

Details

200

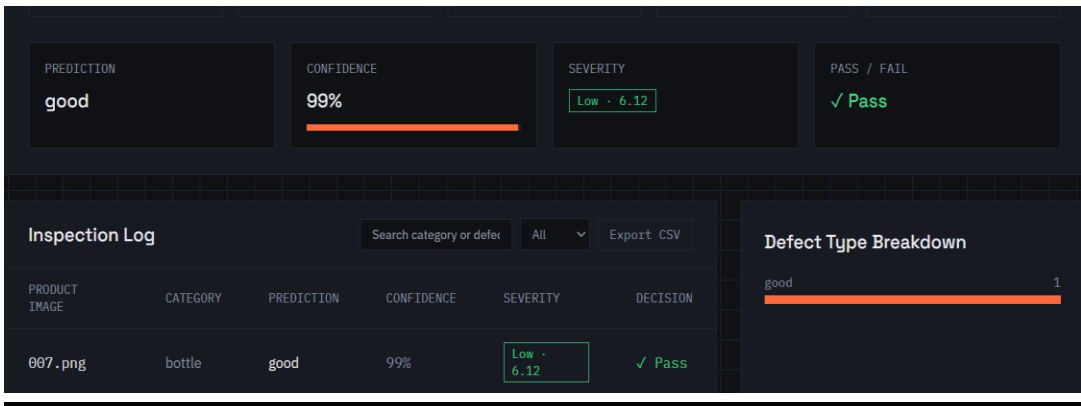
Response body

```
{
  "inspection_id": "81f99eea-4bfd-4b23-a95a-a371a098e381",
  "category": "carpet",
  "is_anomaly": false,
  "defect_result": "PASS",
  "defect_class": "good",
  "confidence_score": 0.99,
  "anomaly_score": 0.12041,
  "threshold": 0.13699,
  "severity_score": 5.64,
  "severity_level": "Low",
  "recommended_action": "Pass - Product is generally acceptable",
  "processing_time_ms": 511.1,
  "yolo_status": "YOLO: Bypassed for category 'carpet'",
  "bbox": [
    0,
    0,
    1024,
    1024
  ],
  "class_probabilities": {
    "good": 99
  },
  "quality_report": {
    "blur_score": 285.42,
    "brightness": 91.73,
    "contrast": 35.58,
    "is_valid": true,
  }
}
```

Response headers

```
access-control-allow-credentials: true
access-control-allow-origin: http://127.0.0.1:8000
content-length: 2922991
content-type: application/json
date: Fri, 24 Jul 2026 10:35:28 GMT
server: uvicorn
vary: Origin
```

Responses



9. Challenges Faced

Several technical challenges were encountered during Milestone 2.

- Designing preprocessing and augmentation APIs suitable for manufacturing datasets.
- Managing large image datasets efficiently.
- Integrating independently developed backend modules.
- Unifying multiple FastAPI services into a single application.
- Replacing frontend mock APIs with real backend endpoints.
- Standardizing API response formats between frontend and backend.
- Supporting multiple manufacturing product categories.
- Displaying AI-generated visualizations directly within the frontend dashboard.

These challenges were addressed through backend refactoring, modular API design, careful integration, and extensive testing.

10. Future Enhancements

The following improvements are planned for future milestones:

- JWT-based authentication
- Database integration for persistent storage
- Cloud deployment
- Real-time webcam inspection
- Batch production monitoring
- Automated PDF report generation
- Role-based access management
- Performance optimization
- Support for additional manufacturing product categories

11. Conclusion

Milestone 2 successfully transformed VisionInspect AI from independently developed components into a fully integrated manufacturing quality inspection platform.

The implemented dataset augmentation, preprocessing services, modular FastAPI backend, AI inspection pipeline, and frontend integration now work together seamlessly to provide real-time defect detection and quality assessment.

The system is capable of performing end-to-end inspection workflows, including image upload, preprocessing, AI inference, visualization, severity estimation, analytics, and reporting. This milestone establishes a strong technical foundation for future deployment and large-scale industrial applications.